Exp No: 5	**Experiments on Packet capture tool: Wireshark**

AIM:

To understand the features of Wireshark as a packet capture tool and to understand the encapsulation of information at the various layers of a protocol stack.

ALGORITHM / PROCEDURE:

1. Install and open Wireshark. The home screen lists all the network interfaces with a small activity graph.
2. Double click the active interface (Ethernet / Wi-Fi) to start the capture.
3. Generate some traffic: open a website in the browser, run `ping google.com` and `nslookup example.com` in the Command Prompt.
4. Stop the capture using the red square button.
5. Apply display filters in the filter bar, for example `icmp`, `dns`, `tcp.port == 80`, `ip.addr == 192.168.1.25`.
6. Select one packet and expand its layers in the **Packet Details** pane. Study the Frame, Ethernet II, Internet Protocol, TCP / ICMP headers.
7. Select any header field and see the corresponding bytes highlighted in the **Packet Bytes** pane.
8. Use **Analyze > Follow > TCP Stream** to follow the full TCP conversation and note the three-way handshake (SYN, SYN-ACK, ACK).
9. Save the capture using **File > Save As (.pcapng)**.

Features of Wireshark

- Captures live packets from Ethernet, Wi-Fi, loopback and other interfaces and also reads saved .pcap files.
- Capture filters (before capture) and display filters (after capture).
- Colouring rules to identify the protocols easily (TCP, UDP, ICMP, ARP, HTTP ...).
- Shows all the protocol headers layer by layer with the raw bytes.
- Statistics tools like Conversations, Protocol Hierarchy and I/O Graph.

Layer	Protocol example	PDU name	Header added
Application	HTTP, DNS	Data / Message	Application header
Transport	TCP, UDP	Segment / Datagram	Port numbers, sequence no. (TCP 20 B, UDP 8 B)
Network	IP, ICMP	Packet	Source and destination IP address (20 B)
Data link	Ethernet	Frame	Source and destination MAC (14 B) + FCS (4 B)
Physical	UTP, Wi-Fi	Bits	None - only signals

DIAGRAM:

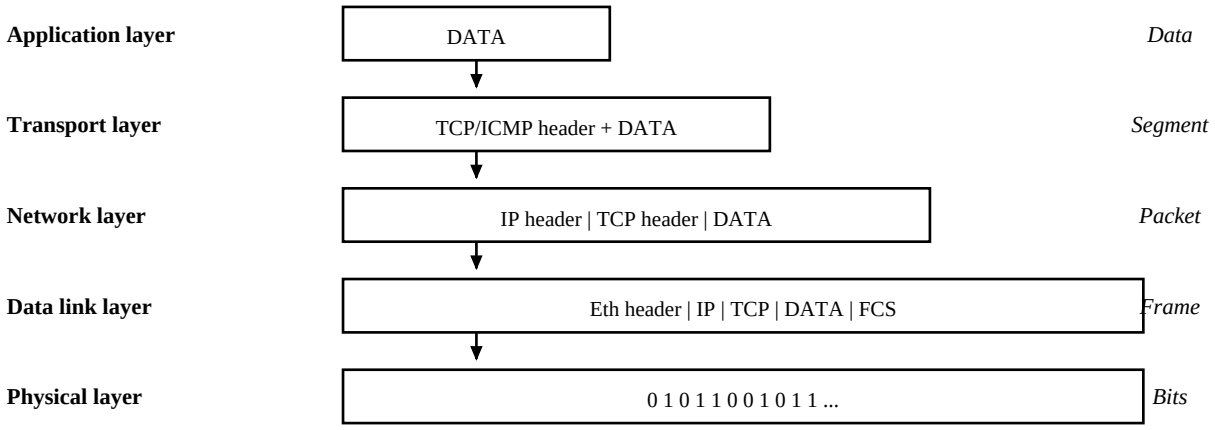

Fig 1: Encapsulation of data as it moves down the protocol stack

OUTPUT:

Capturing from Ethernet - Wireshark

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.1.25	192.168.1.1	DNS	70	Standard query 0x3a1c A example.com
2	0.012431	192.168.1.1	192.168.1.25	DNS	86	Standard query response 0x3a1c A example.com A 93.184.216.34
3	0.013102	192.168.1.25	93.184.216.34	TCP	66	51322 > 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460
4	0.041875	93.184.216.34	192.168.1.25	TCP	66	80 > 51322 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
5	0.041932	192.168.1.25	93.184.216.34	TCP	54	51322 > 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0
6	0.042210	192.168.1.25	93.184.216.34	HTTP	142	GET / HTTP/1.1
7	0.071655	93.184.216.34	192.168.1.25	TCP	54	80 > 51322 [ACK] Seq=1 Ack=89 Win=65535 Len=0
8	0.093340	93.184.216.34	192.168.1.25	HTTP	1294	HTTP/1.1 200 OK (text/html)
9	1.402118	192.168.1.25	142.250.195.78	ICMP	74	Echo (ping) request id=0x0001, seq=21/5376, ttl=128
10	1.416875	142.250.195.78	192.168.1.25	ICMP	74	Echo (ping) reply id=0x0001, seq=21/5376, ttl=117
11	2.118004	c4:e9:84:2a:5b:10	Broadcast	ARP	60	Who has 192.168.1.25? Tell 192.168.1.1

Packet List pane - captured packets (colours show the protocol)

Packet Details - Frame 9 (ICMP Echo request)

```
v Frame 9: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{8F3A21B0}
  , id 0
    Arrival Time: Oct  7, 2026 20:57:38.402118000 India Standard Time
    Frame Length: 74 bytes (592 bits)
    Protocols in frame: eth:ethertype:ip:icmp:data
v Ethernet II, Src: 00:1a:2b:3c:4d:5e, Dst: c4:e9:84:2a:5b:10
    Destination: c4:e9:84:2a:5b:10
    Source: 00:1a:2b:3c:4d:5e
    Type: IPv4 (0x0800)
v Internet Protocol Version 4, Src: 192.168.1.25, Dst: 142.250.195.78
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Total Length: 60
    Identification: 0x7a31 (31281)
    Time to Live: 128
    Protocol: ICMP (1)
    Source Address: 192.168.1.25
    Destination Address: 142.250.195.78
v Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x4d5a [correct]
    Identifier (BE): 1 (0x0001)
    Sequence Number (BE): 21 (0x0015)
    Data (32 bytes)
```

Packet Bytes - Frame 9

```
0000  c4 e9 84 2a 5b 10 00 1a 2b 3c 4d 5e 08 00 45 00   ...*[...+<M^..E.
0010  00 3c 7a 31 00 00 80 01 00 00 c0 a8 01 19 8e fa   .<z1............
0020  c3 4e 08 00 4d 5a 00 01 00 15 61 62 63 64 65 66   .N..MZ....abcdef
0030  67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76   ghijklmnopqrstuv
0040  77 61 62 63 64 65 66 67 68 69                     wabcdefghi
```

Layer	Bytes in the frame	Observed value
Ethernet II	00 - 13 (14 bytes)	`c4:e9:84:2a:5b:10 <- 00:1a:2b:3c:4d:5e, type 0x0800`
IPv4	14 - 33 (20 bytes)	`192.168.1.25 -> 142.250.195.78, TTL 128, protocol 1 (ICMP)`
ICMP	34 - 41 (8 bytes)	`Type 8, Code 0, id 1, seq 21`
Data	42 - 73 (32 bytes)	`abcdefghijklmnopqrstuvwabcdefghi`

RESULT:

Thus the features of Wireshark were studied. Packets were captured and the encapsulation of data at the Data link (Ethernet), Network (IP), Transport (TCP) and ICMP layers was observed in the packet details.